

Arduino Shield – MAX7219 7-Segment Display Driver

PCB Design Project | Tool: Eagle | 2-Layer Shield PCB

Design Tool	Eagle
Board Type	2-Layer Shield PCB
Power Supply	+5V from Arduino Shield Header

1. Project Overview

Arduino shield PCB using MAX7219 SPI 7-segment driver + DS1307 I2C RTC. Displays real-time clock data on 4-digit display. CR1220 battery backup, 32.768kHz crystal. Designed in Eagle CAD.

2. Key Components

- MAX7219CNG – LED Display Driver IC
- DS1307 – Real-Time Clock IC
- Arduino R3 Shield Interface (ICSP + Headers)
- 7-Segment Display – NFD-5641AS (4-digit)
- CR1220 – RTC Backup Battery (SMT)
- 32.768kHz Crystal – RTC clock source
- Pull-up Resistors (4.7k Ω , 27k Ω)
- Decoupling Capacitors (0.1 μ F)

3. Design Features

- MAX7219 drives 4-digit 7-segment display via SPI
- DS1307 RTC provides timekeeping over I2C
- CR1220 battery maintains RTC time during power-off
- Direct plug-in Arduino UNO/Mega shield form factor
- SPI + I2C interfaces used simultaneously
- SMD components for compact shield layout

4. Signal / Functional Flow

- [1] Arduino UNO / Mega – Host MCU ↓
- [2] SPI Interface (MOSI/CLK/CS) ↓
- [3] MAX7219CNG – LED Driver ↓
- [4] NFD-5641AS – 4-Digit 7-Segment Display ↓
- [5] I2C Interface (SDA/SCL) ↓
- [6] DS1307 RTC – Real-Time Clock ↓
- [7] 32.768kHz Crystal ← RTC Clock ↓
- [8] CR1220 Battery ← RTC Backup Power

5. Resume / Portfolio Description

- Designed Arduino shield PCB in Eagle CAD integrating MAX7219 SPI 7-segment driver and DS1307 I2C RTC with CR1220 battery backup.
- Shield features 4-digit 7-segment display, 32.768kHz RTC crystal, and standard Arduino R3 header footprint for direct plug-in compatibility.

6. Short Description (Portfolio / GitHub – ≤350 chars)

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